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Electric cars won't fix sitting: The health costs of designing cities around cars



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Inga Kiderra, UC San Diego



Credit: iStock/ambrozinio

Lawrence Frank quantified that intuition 20 years ago.

In a 2004 study, which was among the first to link neighborhood design and public health, he found that every additional hour a person spent in a car raised the odds of obesity by 6%, while each kilometer walked lowered them by 5. Widely covered by the media, that study helped reframe “car time” for the public as a form of unhealthy sedentary behavior, and it also helped launch a wave of research on how cities shape our well-being.

Now, two decades later, Frank revisits that work in a new commentary in the [American Journal of Preventive Medicine](#). His message is simple: Technological advances like electric vehicles have improved our air quality, but they don’t make us move. We’ve cleaned the tailpipes, but we haven’t cleaned up the behavior.

“You can spend just as much time sitting in an electric vehicle as you can in a gas one,” said Frank, professor of urban studies and planning in the UC San Diego School of Social Sciences. “Every hour in a car will still be a 6% increase in the likelihood of obesity.”

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”

Larry Frank

Frank's original study, part of the Strategies for Metropolitan Atlanta's Transportation and Air Quality (SMARTRAQ) project, broke new ground by pairing GPS-based travel data with health and demographic information. It showed a clear link between community design, time spent in cars, and obesity risk — a finding since replicated worldwide.

Frank's work helped popularize the idea that walkability could be measured and improved, influencing urban planners across North America. It also influenced tools such as Walk Score — familiar to anyone who's ever checked how walkable their own neighborhood might be.

His new commentary, co-authored with Jacob Carson, a student in the UC San Diego Herbert Wertheim School of Public Health and Human Longevity Science-San Diego State University Joint Doctoral Program in Public Health, reviews two decades of evidence showing how built-environment design and the sedentary habits built into everyday life can affect not only obesity but also heart disease, diabetes and even mental health.



Walkable streets and accessible transit help integrate physical activity into daily life. *Credit: Stock/ Oscar Martin*

Walkable, transit-connected communities deliver major health benefits, Frank says. And they deliver the greatest benefits to those who need them most. But most transportation budgets, Frank notes, continue to ignore the health costs of car dependence — costs that, if counted, would make the case for walkable, transit-oriented investment overwhelming.

“We are still driving blind,” he said. “We’re acting like these things don’t exist, but they’re so huge.”

Carson added: “The last 20 years of technological progress haven’t changed the original findings of this paper, and neither will the electrification or autonomization of cars. Robust public transportation and safe cycling and walking infrastructure are still relevant solutions to the challenge of creating healthy and sustainable communities.”

Frank and his collaborators — including his firm Urban Design 4 Health — are developing tools that put hard numbers on those impacts. Their National Public Health Assessment Model lets cities weigh investments in streets, sidewalks and transit using the same cost-benefit frameworks applied to congestion or emissions.

dependence — and integrate active transportation into daily life.”



Design decision shape health at multiple scales. A diagram from a related Cities study shows how regional planning, street networks and pedestrian features influence everyday movement and long-term health. *Credit: UC San Diego*

In a related study appearing in Cities, Frank takes that idea to the street level. The study shows that small-scale features — such as sidewalks, trees, benches, shade and safe street crossings — make a measurable difference, especially for older adults.

“Investing in those small features,” he said, “is fairly easy to do, it’s fairly inexpensive. That’s going to give you the biggest return on investment of anything you can do to improve public health through transportation investment.”

Low-income neighborhoods, he adds, are where those improvements can do the most good.

“Focus the investment on active transportation and pedestrian features,” he said, “in the places where people are most likely to develop chronic disease.”

These are changes cities can make now — without waiting for massive infrastructure bills or having big

“We’ve learned exactly how the built environment shapes behavior,” he said. “Now it’s time to put that knowledge to work — because every mile we walk instead of driving matters.”

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